The muscle to

keep life moving™

About Myotonic Dystrophy Type 1



Onset

Disease symptoms can begin at any point in an affected person's life, depending on the DM1 subtype. Adult-onset DM1 symptoms may appear in the early teens or in middle age.

It is estimated that DM1 affects more than 140,000 people in the United States and over 74,000 people in Europe.

Cause

The underlying cause of DM1 is a defective *DMPK* gene. DM1 is known as a triplet repeat disease because of the expanded number of DNA bases cytosine, thymine, and guanine (CTG), repeating in the *DMPK* gene. These CTG repeats cause toxic RNA to cluster in the nucleus. The toxic RNA binds to RNA splicing proteins, causing altered splicing events in many other genes. This altered splicing results in a wide range of disease symptoms and contributes to the significant disease burden.

Symptoms

Patients experience a broad spectrum of symptoms, including:

- Muscle weakness throughout the body, from difficulties with facial and swallowing muscles to hand and grip weakness to lower leg and foot weakness, foot drop and resulting mobility issues
- Myotonia, or difficulty relaxing muscles in the jaw, tongue, throat muscles, lower arm and hand
- Respiratory issues
- Gastrointestinal dysfunction
- Cognitive deficits, excessive daytime sleepiness and dysregulated sleep
- Early cataracts

Our goal: Targeting the genetic basis of the disease

Our DM1 program candidates consist of:



A proprietary antigen-binding fragment (Fab), which is the portion of an antibody that binds to antigens



Connected with a clinically validated linker

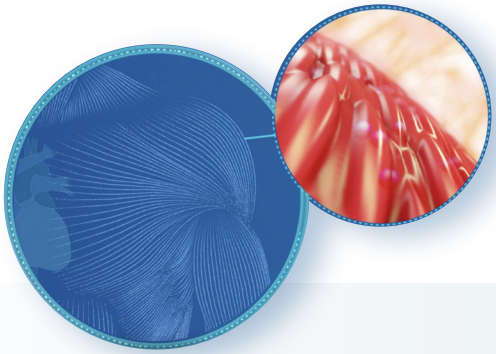


Connected to an antisense oligonucleotide (ASO)

Our program candidates are designed to address the genetic basis of DM1 by reducing the levels of mutant *DMPK* RNA in the nucleus. Knocking down this toxic RNA is expected to release splicing proteins and enable normal mRNA processing, potentially halting or reversing progression of the disease. In preclinical studies, we have observed reduction of nuclear foci and correction of splicing in DM1 patient cells, robust reduction in toxic human nuclear *DMPK* RNA at four weeks in a novel *in vivo* model developed by Dyne, reversal of myotonia after a single dose in a DM1 disease model, durability of response up to 12 weeks and enhanced muscle distribution.

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Our approach: The FORCE™ platform

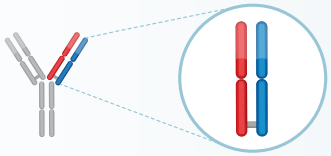


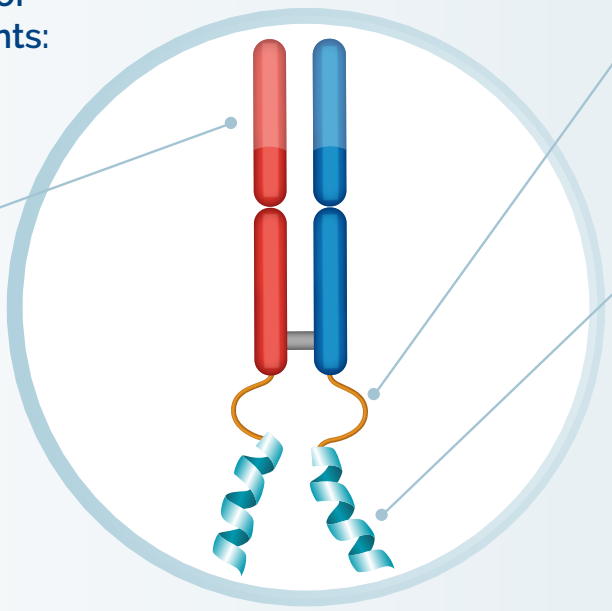
Our proprietary FORCE™ platform drives our efforts to develop and deliver targeted, modern oligonucleotide therapies that are designed to overcome limitations in delivery to muscle tissue seen with other approaches.

Our therapeutics consist of three essential components:

1. Antibody

Proprietary Fab targets TfR1 to enable muscle delivery



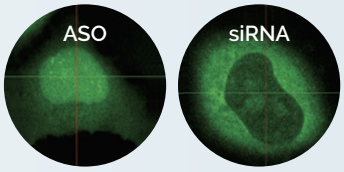


2. Linker

Clinically validated, enables precise conjugation of multiple payloads to a single Fab

3. Payload

Modularity enables rational selection of payload to target the genetic basis of disease



Nuclear localization

Cytoplasmic localization

Adapted from: Ohrt et al: *NAR*, 2006, v.34, p.1369

We believe our FORCE™ platform provides several advantages, including:

- Targeted delivery to muscle tissue
- Extended time between doses
- Ability to re-dose
- Ability to target the genetic basis of disease stop/reverse progression

To learn more about our FORCE™ platform and DM1 program visit our website: www.dyne-tx.com.

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